

KuduDash

Test Plan & Results

Backend Unit Testing | Jest | May 2026

1. Testing Strategy

KuduDash adopted a unit testing strategy for its backend, using Jest as the testing framework. All controller logic and route handlers were tested in isolation by mocking database calls (MongoDB/Mongoose) and external dependencies such as Auth0 and the PayFast payment gateway. Tests were written alongside feature development across the four sprints, with each controller having a dedicated test file under `src/__tests__`.

The testing focus covered four main categories:

- Happy path — verifying that valid inputs produce the correct response and data.
- Error handling — ensuring 400, 403, 404, and 500 HTTP responses are returned appropriately for edge cases.
- Business rule enforcement — e.g. preventing duplicate menu items, blocking suspended vendors, validating payment signatures.
- Data integrity — confirming that database save and update operations are called with the correct data.

2. Test Summary

Test Suites	Tests Passed	Tests Skipped	Total Tests
13	204	19	223

All 204 active tests passed. The 19 skipped tests belong to the `analyticsController` suite, which was deferred as the analytics feature uses aggregated read-only queries that are covered by integration-level validation in the live environment.

3. Test Cases

The following table lists all test cases organised by test suite and controller group.

Test Suite	Controller / Group	Test Case	Result
MenuItem Controller	<code>getMenuItems</code>	returns available menu items for a vendor	Pass
		returns 500 on database error	Pass
	<code>getMenuItemById</code>	returns a menu item when found	Pass
		returns 404 if item not found	Pass
		returns 500 on database error	Pass
	<code>getAllMenuItemsByVendor</code>	returns all menu items (available and unavailable) for a vendor	Pass

Test Suite	Controller / Group	Test Case	Result
		returns empty array if vendor has no menu items	Pass
		returns 500 on database error	Pass
	<i>createMenuItem</i>	creates a menu item when no duplicate exists	Pass
		returns 409 if duplicate name exists (case-insensitive)	Pass
		returns 409 on Mongo duplicate key error	Pass
		returns 400 on validation error	Pass
	<i>updateMenuItem</i>	updates menu item successfully	Pass
		returns 404 if item does not exist before update	Pass
		returns 409 if duplicate name exists	Pass
		returns 404 if item not found during update (no name in body)	Pass
		returns 500 on database error	Pass
	<i>deleteMenuItem</i>	deletes menu item successfully	Pass
		returns 404 if item not found	Pass
		returns 500 on database error	Pass
Menu Item Availability	<i>toggleAvailability</i>	marks an available item as sold out	Pass
		restores a sold-out item to available	Pass
		calls save on the item to persist availability to the database	Pass
		returns the updated isSoldOut value in the response	Pass
		returns 404 if menu item is not found	Pass
		returns 403 if vendor does not own the item	Pass
		returns 500 on database error	Pass
	<i>getMenuItems (student-facing — available items only)</i>	returns only available items for a vendor	Pass
		returns an empty array when all items are sold out	Pass
		returns 500 on database error	Pass
	<i>getAllMenuItemsByVendor (vendor dashboard)</i>	returns all items including sold-out ones for the vendor dashboard	Pass
		returns 500 on database error	Pass

Test Suite	Controller / Group	Test Case	Result
Menu Item Allergen & Dietary Tags	<i>createMenuitem — with allergen and dietary tags</i>	creates a menu item with dietary labels	Pass
		creates a menu item with allergen warnings	Pass
		creates a menu item with no dietary or allergen tags	Pass
		returns 409 if duplicate item name exists	Pass
		returns 500 on database error	Pass
	<i>updateMenuitem — allergen and dietary tags</i>	updates dietary labels on an existing menu item	Pass
		updates allergen warnings on an existing menu item	Pass
		returns the updated item with saved tags	Pass
		returns 404 if item not found	Pass
		returns 500 on database error	Pass
	<i>getMenuitems — dietary filter (student-facing)</i>	returns items with their dietary labels and allergens	Pass
		returns only items matching a dietary filter	Pass
		returns an empty array when no items match the dietary filter	Pass
		returns 500 on database error	Pass
		returns full allergen and dietary detail for a single item	Pass
Cart Controller	<i>getCart</i>	returns item with empty allergen and dietary arrays when none are set	Pass
		returns 404 if item not found	Pass
		returns 500 on database error	Pass
	<i>addItem</i>	returns empty cart if none exists	Pass
		returns cart if found	Pass
		handles server error	Pass
		returns 400 if required fields missing	Pass
		creates new cart and adds item	Pass
	<i>updateItemQuantity</i>	increments quantity if item already exists	Pass
		handles server error	Pass
		returns 404 if cart not found	Pass
		returns 404 if item not found in cart	Pass
		updates item quantity	Pass

Test Suite	Controller / Group	Test Case	Result
		removes item if quantity <= 0	Pass
	<i>removeItem</i>	returns 404 if cart not found	Pass
		removes item from cart	Pass
		handles server error	Pass
	<i>clearCart</i>	clears the cart	Pass
		handles server error	Pass
Vendor Controller	<i>getAllVendors</i>	returns active vendors	Pass
		handles server error	Pass
	<i>getVendorById</i>	returns vendor if found	Pass
		returns 404 if vendor not found	Pass
		handles server error	Pass
	<i>createVendor</i>	creates a vendor	Pass
		handles validation error	Pass
	<i>updateVendor</i>	updates vendor	Pass
		returns 404 if vendor not found	Pass
		handles server error	Pass
	<i>updateVendorProfile</i>	updates allowed fields and logo	Pass
		returns 404 if vendor not found	Pass
		handles server error	Pass
	<i>suspendVendor</i>	suspends a vendor with a reason	Pass
		returns 400 if reason is missing	Pass
		returns 400 if reason is blank	Pass
		returns 404 if vendor not found	Pass
		handles server error	Pass
	<i>reinstateVendor</i>	reinstates a vendor	Pass
		returns 404 if vendor not found	Pass
		handles server error	Pass
	<i>requireNotSuspended</i>	calls next() if no vendorId is present	Pass
		calls next() if vendor is active	Pass
		returns 403 if vendor is suspended	Pass
		returns 404 if vendor not found	Pass
		handles server error	Pass

Test Suite	Controller / Group	Test Case	Result
Payment Controller	<i>initiatePayment</i>	returns pfData and pfUrl for a valid order	Pass
		returns 404 if order not found	Pass
		returns 403 if student does not own the order	Pass
		returns 400 if order is already paid	Pass
		returns 500 on server error	Pass
	<i>handleNotify</i>	returns 400 if signature is invalid	Pass
		marks order as paid on COMPLETE payment with valid signature	Pass
		does not update order if already paid	Pass
		does nothing if payment_status is not COMPLETE	Pass
		returns 500 on server error	Pass
	<i>verifyPayment</i>	returns order status if already paid	Pass
		promotes order from received to paid (ITN fallback)	Pass
		returns 404 if order not found	Pass
		returns 403 if student does not own the order	Pass
		returns 500 on server error	Pass
Review Controller	<i>submitReview</i>	should return 404 if order not found	Pass
		should return 403 if order does not belong to student	Pass
		should return 400 if order has not been collected yet	Pass
		should return 400 if order has already been reviewed	Pass
		should create and return a review for a collected order	Pass
		should return 500 on unexpected error	Pass
	<i>getVendorReviews</i>	should return reviews for a vendor	Pass
		should return empty array if vendor has no reviews	Pass
		should return 500 on error	Pass
	<i>getReviewByOrder</i>	should return the review for a given order	Pass
		should return null if no review exists for the order	Pass
		should return 500 on error	Pass
Order Controller	<i>getOrders</i>	should return orders for a student	Pass

Test Suite	Controller / Group	Test Case	Result
		should return 500 on error	Pass
	<i>getOrderByld</i>	should return an order by ID	Pass
		should return 404 if order not found	Pass
		should return 500 on error	Pass
	<i>createOrder</i>	should create and return a new order	Pass
		should return 400 on error	Pass
	<i>updateOrderStatus</i>	should update and return the order	Pass
		should return 404 if order not found	Pass
		should return 500 on error	Pass
	<i>getVendorOrders</i>	should return all orders for a vendor	Pass
		should filter by status if provided	Pass
		should return 500 on error	Pass
	<i>getAllOrders</i>	should return all orders	Pass
		should return 500 on error	Pass
	<i>getOrdersByStudent</i>	should return orders for a student	Pass
		should return 200 with message if no orders found	Pass
		should return 500 on error	Pass
	<i>getOrdersByVendor</i>	should return orders for a vendor	Pass
		should return 200 with message if no orders found	Pass
		should return 500 on error	Pass
	<i>updateOrderStatus (additional branches)</i>	should set collectionCode when status is ready	Pass
		should include estimatedReadyAt in update when provided	Pass
Admin Controller	<i>getAllAdmins</i>	should return 400 for an invalid status	Pass
		returns all admins	Pass
		handles server error	Pass
	<i>getAdminByld</i>	returns admin if found by authProviderId	Pass
		returns 404 if admin not found	Pass
		handles server error	Pass
	<i>createAdmin</i>	creates an admin	Pass
		handles validation error	Pass

Test Suite	Controller / Group	Test Case	Result
	<i>updateAdmin</i>	updates admin	Pass
		returns 404 if admin not found	Pass
		handles server error	Pass
Appeal Controller	<i>submitAppeal</i>	creates an appeal when vendor is suspended and message is provided	Pass
		returns 400 if message body is missing	Pass
		returns 404 if vendor not found	Pass
		returns 400 if vendor is not suspended	Pass
		returns 400 if vendor already has a pending appeal	Pass
		returns 500 on database error	Pass
	<i>getAllAppeals</i>	returns all appeals	Pass
		returns an empty array when no appeals exist	Pass
		returns 500 on database error	Pass
	<i>markReviewed</i>	marks an appeal as reviewed with admin feedback	Pass
		returns 404 if appeal not found	Pass
		returns 500 on database error	Pass
	<i>getVendorAppeal</i>	returns the vendor's most recent appeal	Pass
		returns 404 if no appeal found for vendor	Pass
		returns 500 on database error	Pass
	<i>reinstateVendor</i>	reinstates the vendor and sets their status to active	Pass
		returns 404 if vendor not found	Pass
		returns 500 on database error	Pass
Auth Routes	<i>POST /sync</i>	returns existing student	Pass
		returns new user	Pass
		handles DB error	Pass
		returns existing vendor with vendor fields	Pass
		returns existing admin with admin fields	Pass
		backfills authProviderId when matched by email but sub differs	Pass
	<i>POST /register/student</i>	creates student	Pass
		409 if exists	Pass

Test Suite	Controller / Group	Test Case	Result
		returns 409 on duplicate key error	Pass
		returns 500 on unexpected error	Pass
	POST /register/vendor	requires business name	Pass
		creates vendor	Pass
		returns 409 if vendor already exists	Pass
		returns 409 on duplicate key error	Pass
		returns 500 on unexpected error	Pass
	POST /register/admin	rejects wrong code	Pass
		creates admin	Pass
		returns 400 if adminCode is missing	Pass
		returns 409 if admin already exists	Pass
		returns 409 on duplicate key error	Pass
		returns 500 on unexpected error	Pass
Student Controller	getAllStudents	returns all students	Pass
		handles server error	Pass
	getStudentProfile	returns student profile if found	Pass
		returns 404 if student not found	Pass
		handles server error	Pass
	createStudent	creates a student	Pass
		handles validation error	Pass
	updateStudent	updates student	Pass
		returns 404 if student not found	Pass
		handles server error	Pass
Codecov Smoke Test	Utility	adds two numbers	Pass
		returns true for a non-empty string	Pass

4. Test Results

Tests were executed using the Jest test runner with the --verbose flag. The full output is included below, captured directly from the terminal on 19 May 2026.

Test File	Suite	Passed	Skipped	Status
<i>menuItemController.test.js</i>	MenuItemController	20	0	Pass

Test File	Suite	Passed	Skipped	Status
<i>menuItemAvailability.test.js</i>	Menu Item Availability	12	0	Pass
<i>menuItemAllergens.test.js</i>	Menu Item Allergen & Dietary Tags	18	0	Pass
<i>cartController.test.js</i>	Cart Controller	17	0	Pass
<i>vendorController.test.js</i>	Vendor Controller	26	0	Pass
<i>paymentController.test.js</i>	Payment Controller	15	0	Pass
<i>Reviewcontroller.test.js</i>	Review Controller	12	0	Pass
<i>orderController.test.js</i>	Order Controller	24	0	Pass
<i>adminController.test.js</i>	Admin Controller	10	0	Pass
<i>appealController.test.js</i>	Appeal Controller	18	0	Pass
<i>auth.routes.test.js</i>	Auth Routes	23	0	Pass
<i>studentController.test.js</i>	Student Controller	10	0	Pass
<i>sample.test.js</i>	Codecov Smoke Test	2	0	Pass
<i>analyticsController.test.js</i>	Analytics Controller	0	19	Skipped
TOTAL	13 suites	204	19	204/204

Jest run completed in 4.717 seconds (estimated 6 s). All 204 active tests passed with 0 failures.